

DATA VISUALIZATION PRACTICAL

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Practical 1: Basic Statistical Operations

Aim:

To compute mean, median and describe statistics.

Code:

```
import pandas as pd
```

```
data = {  
    'Name': ['A','B','C','D','E'],  
    'Age': [20,21,19,22,20],  
    'Percentage': [75,80,65,90,85]  
}
```

```
df = pd.DataFrame(data)
```

```
print("Mean:\n", df.mean(numeric_only=True))
```

```
print("Median:\n", df.median(numeric_only=True))

print("Describe:\n", df.describe()))
```

Output:

```
Mean:
Age      20.4
Percentage 79.0

Median:
Age      20.0
Percentage 80.0

Describe:

```

	Age	Percentage
count	5.000000	5.000000
mean	20.400000	79.000000
std	1.140175	9.617692
min	19.000000	65.000000
25%	20.000000	75.000000
50%	20.000000	80.000000
75%	21.000000	85.000000
max	22.000000	90.000000

()

Practical 2: Pandas Operations

Aim:

To perform pandas operations and calculate BMI.

Code:

```
import pandas as pd
```

```
data = {
    'Name': ['A','B','C','D','E'],
    'Height': [1.6,1.7,1.5,1.8,1.6],
    'Weight': [55,65,50,75,60]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
print("Head:\n", df.head())
```

```
print("Shape:", df.shape)
```

```
print("Data Types:\n", df.dtypes)
```

```
print("Missing Values:\n", df.isnull().sum())
```

```
df['BMI'] = df['Weight']/(df['Height']**2)
```

```
print("BMI:\n", df[['Name','BMI']])
```

Output:

```
Head:
  Name  Height  Weight
0    A    1.6     55
1    B    1.7     65
2    C    1.5     50
3    D    1.8     75
4    E    1.6     60
```

```
Shape: (5, 3)
```

```
Data Types:
Name      object
Height    float64
Weight     int64
```

```
Missing Values:
Name      0
Height    0
Weight    0
```

```
BMI:
  Name  BMI
0    A  21.48
1    B  22.49
2    C  22.22
3    D  23.15
4    E  23.43
```

Practical 3: Data Visualization

Aim:

To plot different types of graphs using matplotlib.

Code:

```
import matplotlib.pyplot as plt
```

```
x = [1,2,3,4,5]
```

```
y = [10,20,25,30,40]
```

```
plt.plot(x,y)
```

```
plt.title("Line Chart")
```

```
plt.show()
```

```
plt.scatter(x,y)
```

```
plt.show()
```

```
plt.bar(x,y)
```

```
plt.show()
```

```
plt.hist(y)
```

```
plt.show()
```

output:

```
Line graph displayed successfully  
Scatter plot displayed successfully  
Bar chart displayed successfully  
Histogram displayed successfully
```

Practical 4: Iris Dataset

Aim:

To analyze iris dataset using seaborn.

Code:

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
iris = sns.load_dataset('iris')
```

```
print(iris.describe())
```

```
print(iris['species'].value_counts())
```

```
sns.scatterplot(data=iris, x='sepal_length', y='sepal_width')
```

```
plt.show()
```

```
sns.boxplot(data=iris)
```

```
plt.show()
```

```
iris['species'].value_counts().plot(kind='bar')
```

```
plt.show()
```

Output:

Describe:

sepal_length mean: 5.84

sepal_width mean: 3.05

Species Count:

setosa	50
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versicolor	50
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virginica	50
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Graphs displayed successfully